Major Vessels of the Neck

1. The right common carotid artery arises from the brachiocephalic trunk behind the right sternoclavicular joint. From which precise anatomical structure does the left common carotid artery directly originate?
a. Brachiocephalic trunk
b. Left subclavian artery
c. Internal thoracic artery
d. Arch of the aorta
e. Left pulmonary artery
2. The common carotid artery ascends within the carotid sheath and bifurcates into the internal and external carotid arteries. This high-yield surgical landmark normally occurs at which level?
a. Lower border of the cricoid cartilage
b. Jugular notch of the sternum
c. Upper border of the thyroid cartilage (C4 vertebra level)
d. Angle of the mandible
e. Hyoid bone greater horn
3. The external carotid artery is the chief arterial supply to the face and scalp. How many total anatomical branches does it give off in the neck region?
a. Eight branches
b. Six branches
c. Four branches
d. Five branches
e. Three branches
4. The first anterior branch of the external carotid artery arises immediately above the carotid bifurcation and runs downwards and forwards to supply the upper pole of the thyroid gland. This is the:
a. Lingual artery
b. Facial artery
c. Ascending pharyngeal artery
d. Superior thyroid artery
e. Occipital artery
5. The superior thyroid artery runs in close proximity to a nerve near the upper pole of the thyroid gland. To avoid injuring this nerve, which supplies the cricothyroid muscle and alters vocal pitch, the superior thyroid artery must be ligated as close to the gland as possible. This nerve is the:
a. Internal laryngeal nerve
b. External laryngeal nerve
c. Recurrent laryngeal nerve
d. Descendens hypoglossi
e. Mylohyoid nerve
6. The second anterior branch of the external carotid artery arises at the level of the greater cornu of the hyoid bone. It forms a characteristic loop directed upwards and forwards, which is crossed superficially by the hypoglossal nerve (CN XII). This vessel is the:
a. Lingual artery

b. Facial artery
c. Maxillary artery
d. Ascending pharyngeal artery
e. Superior laryngeal artery

7. The lingual artery runs deep to which muscle of the tongue, dividing it into three distinct anatomical parts for descriptive surgical purposes?

a. Genioglossus
b. Styloglossus
c. Hyoglossus muscle
d. Palatoglossus
e. Mylohyoid

8. The third anterior branch of the external carotid artery arises just above the lingual artery, passes deep to the posterior belly of the digastric, and exhibits a highly tortuous loop that grooves the posterior border of the submandibular salivary gland. This is the:

a. Maxillary artery
b. Occipital artery
c. Posterior auricular artery
d. Facial artery
e. Ascending palatine artery

9. As the facial artery loops over the lower border of the mandible to enter the face, it runs immediately anterior to which structural muscle insert?

a. Buccinator muscle
b. Medial pterygoid muscle
c. Masseter muscle (anterior border)
d. Platysma sheet
e. Lateral pterygoid muscle

10. The only medial branch of the external carotid artery is a small, slender vessel that ascends vertically on the lateral wall of the pharynx deep to the internal carotid artery. This branch is the:

a. Superior pharyngeal artery
b. Ascending pharyngeal artery
c. Deep cervical artery
d. Palatine branch
e. Ascending palatine trunk

11. The external carotid artery terminates behind the neck of the mandibular condyle within the parenchyma of the parotid salivary gland by dividing into which two major vessels?

a. Maxillary and superficial temporal arteries
b. Facial and lingual arteries
c. Posterior auricular and occipital arteries
d. Middle meningeal and deep temporal arteries
e. Transverse facial and infraorbital arteries

12. The internal carotid artery ascends from the common carotid bifurcation to enter the skull base. Which unique feature characterizes its entire course within the neck region?

a. It gives off three small muscular branches to the strap muscles
b. It gives off zero (none) branches in the neck

c. It gives off the middle meningeal artery near the hyoid bone
d. It branches extensively into the prevertebral muscles
e. It gives off the ophthalmic artery before reaching the temporal bone

13. At its origin, the internal carotid artery exhibits a localized dilation containing baroreceptors that monitor systemic blood pressure variations. This dilatation is the:

a. Carotid body
b. Carotid glomus
c. Carotid sinus
d. Jugular bulb
e. Aortic arch sinus

14. The subclavian artery is divided into three functional parts by its relationship to a key muscular landmark in the root of the neck. This landmark muscle is the:

a. Scalenus posterior
b. Scalenus anterior muscle
c. Scalenus medius
d. Sternocleidomastoid
e. Longus colli

15. The first part of the subclavian artery gives off a major branch that ascends vertically to pass through the foramina transversaria of vertebrae C1 through C6. This critical vessel is the:

a. Internal thoracic artery
b. Thyrocervical trunk
c. Costocervical trunk
d. Vertebral artery
e. Ascending cervical artery

16. The thyrocervical trunk is a short branch arising from the first part of the subclavian artery. It immediately splits into three high-yield vessels, which are the:

a. Superior thyroid, lingual, and facial arteries
b. Inferior thyroid, suprascapular, and transverse cervical arteries
c. Vertebral, internal thoracic, and deep cervical arteries
d. Maxillary, mandibular, and incisive arteries
e. Occipital, posterior auricular, and ascending pharyngeal arteries

17. The inferior thyroid artery arches medially behind the carotid sheath to supply the lower pole of the thyroid gland. During thyroid surgery, it must be ligated as far away from the gland as possible to prevent iatrogenic damage to which intimately related nerve?

a. External laryngeal nerve
b. Internal laryngeal nerve
c. Recurrent laryngeal nerve
d. Phrenic nerve
e. Ansa cervicalis root

18. Let's trace the high-yield venous unions ($X + Y = Z$). The superficial temporal vein joins the maxillary vein behind the neck of the mandible to form which major vascular trunk?

a. Internal jugular vein
b. External jugular vein
c. Retromandibular vein
d. Common facial vein

e. Anterior jugular vein

19. Within the lower part of the parotid gland, the retromandibular vein splits into anterior and posterior divisions. The posterior division joins the posterior auricular vein to directly form the:

a. Internal jugular vein
b. External jugular vein
c. Anterior jugular vein
d. Facial vein trunk
e. Subclavian vein channel

20. The external jugular vein runs vertically down the neck over the superficial surface of the sternocleidomastoid muscle, deep to the platysma, and terminates by draining directly into the:

a. Internal jugular vein
b. Subclavian vein
c. Brachiocephalic vein
d. Anterior jugular arch
e. Retromandibular confluence

21. Let's trace another union. The anterior division of the retromandibular vein emerges from the parotid gland and joins the facial vein to form which high-yield vascular channel?

a. External jugular vein
b. Common facial vein
c. Anterior jugular vein
d. Internal jugular vein terminal
e. Lingual venous arch

22. The common facial vein runs downwards and backwards across the carotid triangle to empty its contents directly into the:

a. Subclavian vein
b. External jugular vein
c. Internal jugular vein
d. Brachiocephalic trunk
e. Anterior jugular venous arch

23. The anterior jugular veins descend near the midline of the neck. Just above the sternum inside the suprasternal space of Burns, the right and left anterior jugular veins are joined horizontally by a distinct vessel called the:

a. Pterygoid venous arch
b. Jugular venous arch
c. Cavernous connection
d. Subclavian bridge
e. Common facial anastomosis

24. The internal jugular vein is a direct continuous mapping of which intracranial venous sinus after it exits the skull base via the jugular foramen?

a. Superior sagittal sinus
b. Cavernous sinus
c. Sigmoid sinus (at the superior bulb of the internal jugular vein)
d. Inferior petrosal sinus purely
e. Straight sinus

25. Throughout its entire course down the neck, the internal jugular vein runs inside the carotid sheath, maintaining which relative position to the common carotid artery?
a. Immediately anterior to the artery
b. Medial to the artery
c. Lateral to the artery
d. Deep or posterior to the artery
e. It completely wraps around the artery like a sleeve
26. At the root of the neck, the internal jugular vein joins the subclavian vein to form which massive venous channel heading toward the superior vena cava?
a. Superior intercostal vein
b. Axillary vein
c. Brachiocephalic vein (innominate vein)
d. Internal thoracic vein
e. Azygos vein system
27. The thoracic duct is the largest lymphatic channel in the body. It ascends into the left root of the neck, forms a dramatic arch arching laterally behind the carotid sheath, and empties its contents into the junction between which two major veins?
a. Left internal jugular and left subclavian veins (the left venous angle)
b. Right common facial and right internal jugular veins
c. Left external jugular and left subclavian veins
d. Right brachiocephalic and right subclavian veins
e. Left axillary and left cephalic veins
28. Winding downwards and forwards across the superficial face of the scalenus anterior muscle, passing directly between the subclavian artery and subclavian vein, is which vital motor nerve?
a. Vagus nerve (CN X)
b. Sympathetic trunk chain
c. Spinal accessory nerve (CN XI)
d. Phrenic nerve (derived from C3, C4, C5)
e. Long thoracic nerve
29. The vagus nerve (CN X) maintains a highly precise, constant relationship to the major vessels within the carotid sheath. Throughout its course in the neck, it sits:
a. Directly anterior to the common carotid artery
b. Posteriorly in the groove between the carotid artery and internal jugular vein
c. Lateral to the internal jugular vein
d. Superficial to the internal jugular vein
e. Inside the lumen of the internal jugular vein
30. A rise in intracranial pressure or severe packing of deep nodes can compress the internal jugular vein. This venous system contains a localized bicuspid valve situated at which location to prevent the retrograde backflow of blood toward the brain?
a. At the superior bulb within the jugular foramen
b. Midway at the level of the hyoid bone
c. At the inferior bulb near its termination into the brachiocephalic vein
d. Inside the common facial vein junction
e. The internal jugular vein is completely valveless throughout life